Scalability & Future Plan

Date	29 October 2026
Team ID	N/A - individual project
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Synthetic dataset	Insights are not validated market facts.	High
2	No measured performance baseline	Render/query/filters timing absent; no load test data.	Medium
3	Static CSV / SQLite prototype	Updates require data refresh, DB rebuild and Tableau republish.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution	
1	User load	Single deployed student site; no load test.	Use production-grade hosting, caching and monitoring.	Medium
2	Data storage	SQLite DB sized for supplied data; file-based update.	Move to managed relational DB for concurrent writes/large growth.	Medium
3	Performance	Visuals and endpoints tested locally as reported; timings not measured.	Record render and filter timings; optimize extracts, indexes and queries.	High
4	Security	Public Tableau views and website; local .env is excluded from repo.	Keep secrets in host environment; add access controls for any private data.	High

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact

Phase 2	Document field dictionary and source provenance	After project submission	More transparent data definitions.
Phase 3	Run formal usability/performance tests	Future iteration	Evidence-based improvement priorities.
Phase 4	Add refreshed validated data / managed storage if approved	Future iteration	Improved relevance and maintainability.